



IBM Spectrum Storage

IBM Spectrum Virtualize

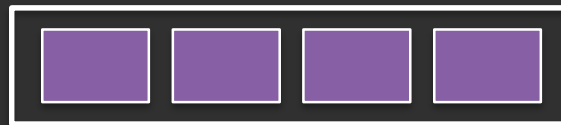
Hot Spare Node

Chris Bulmer, *North America User Group, June 2016*

IBM Confidential









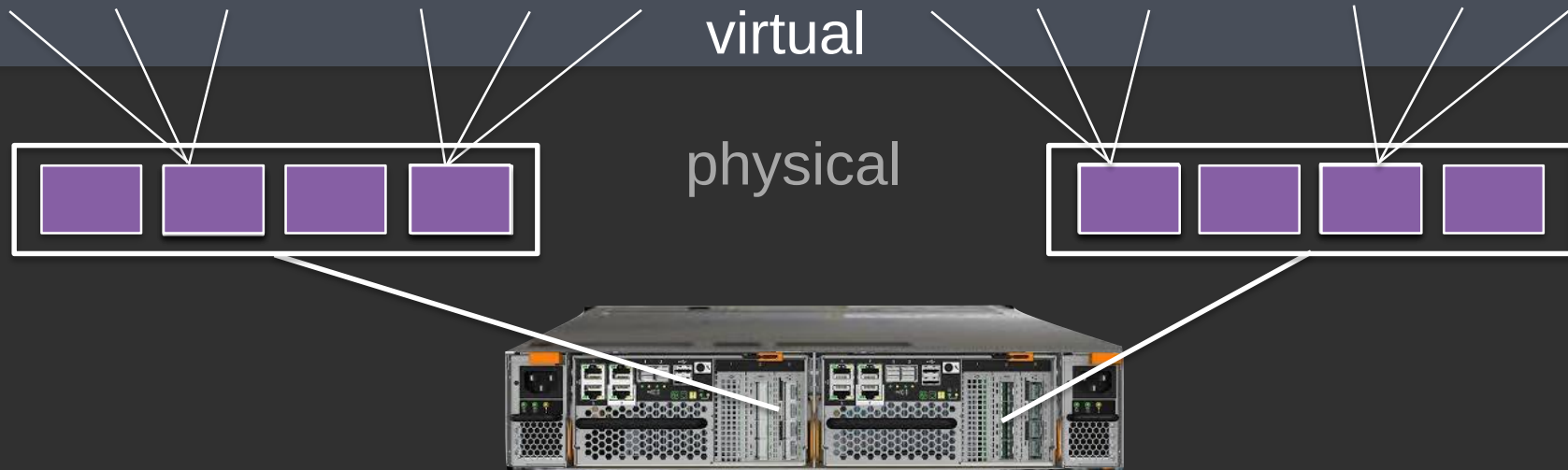
Hot Spare Node NPIV



host port
primary port
failover host port

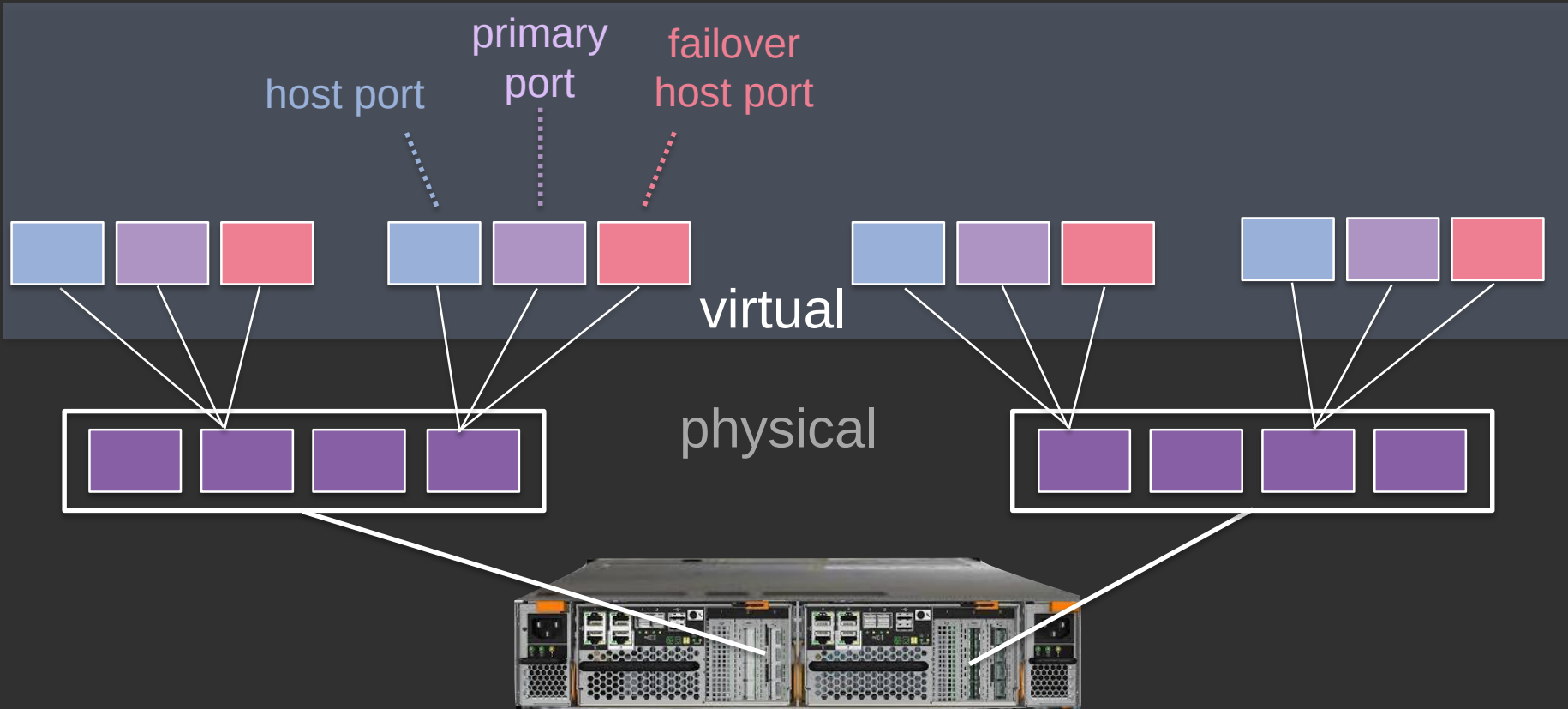
virtual

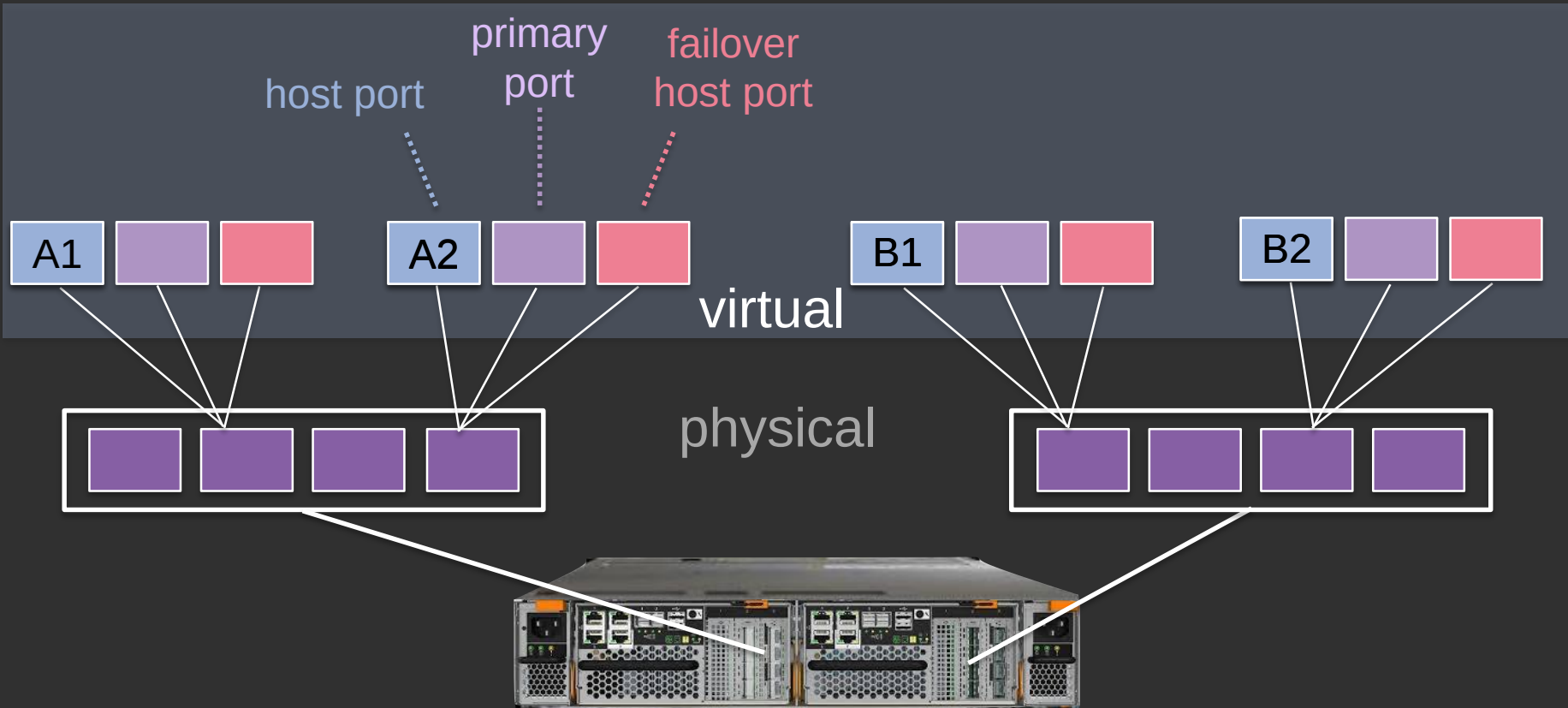
physical

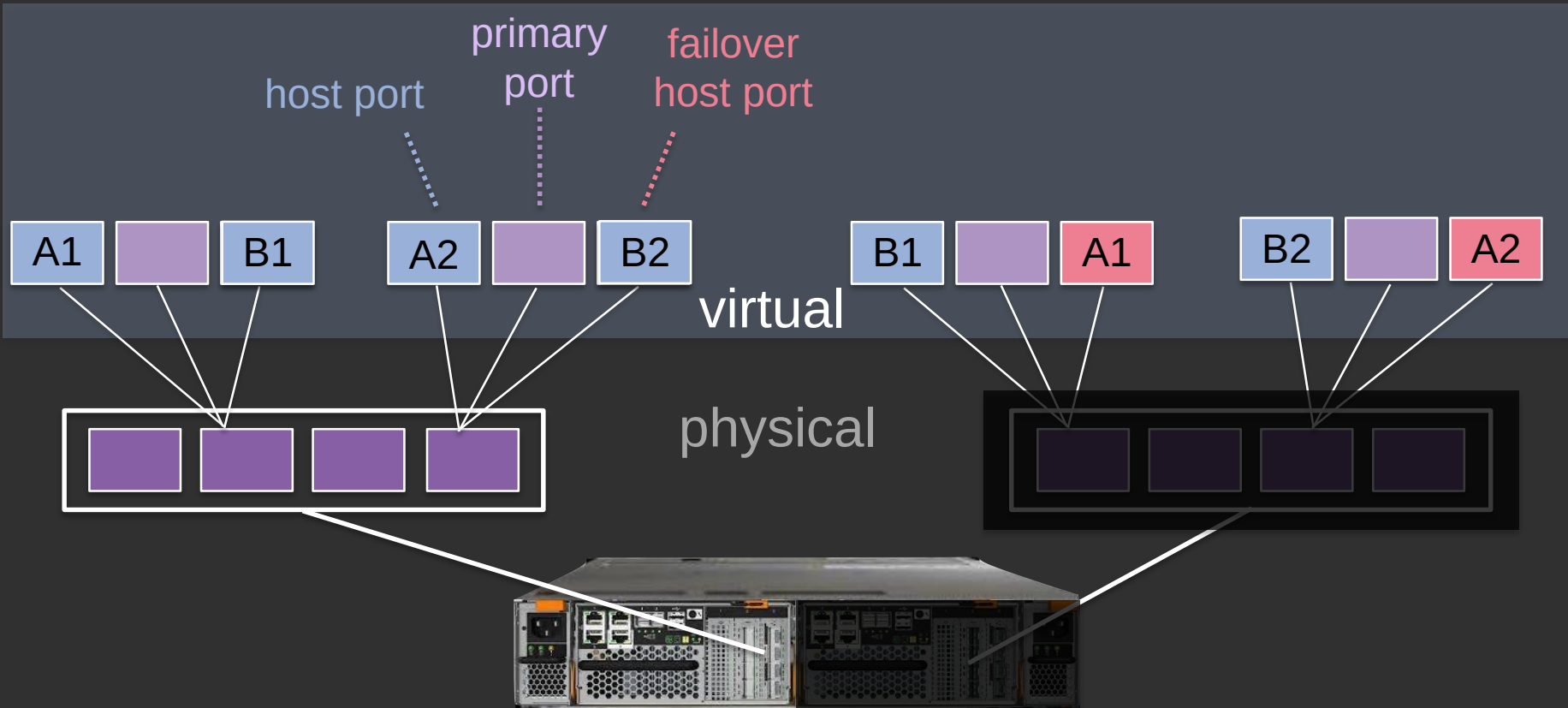




Hot Spare Node NPIV





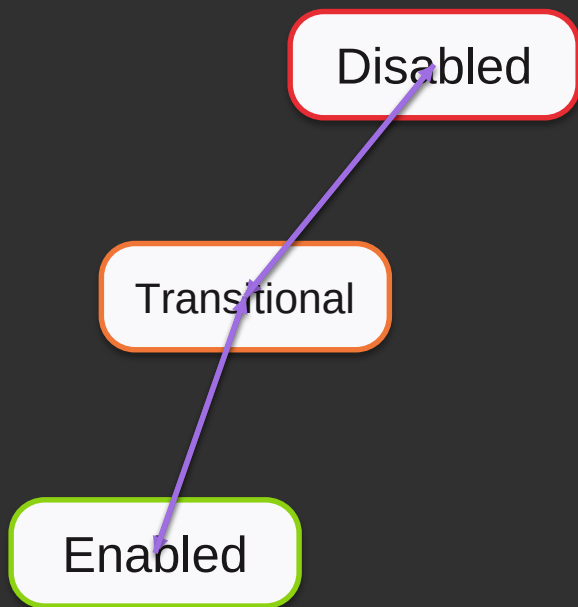




- Host ports are **target only** ports designed only for host traffic
- Primary ports are used for backend and inter-node (local and remote) traffic
- Unique WWPN
- Same WWNN as “owning” node
- Enabled only when node is ready for IO after startup
- Note: rmnode command will unconfigure host ports so port failover will not occur



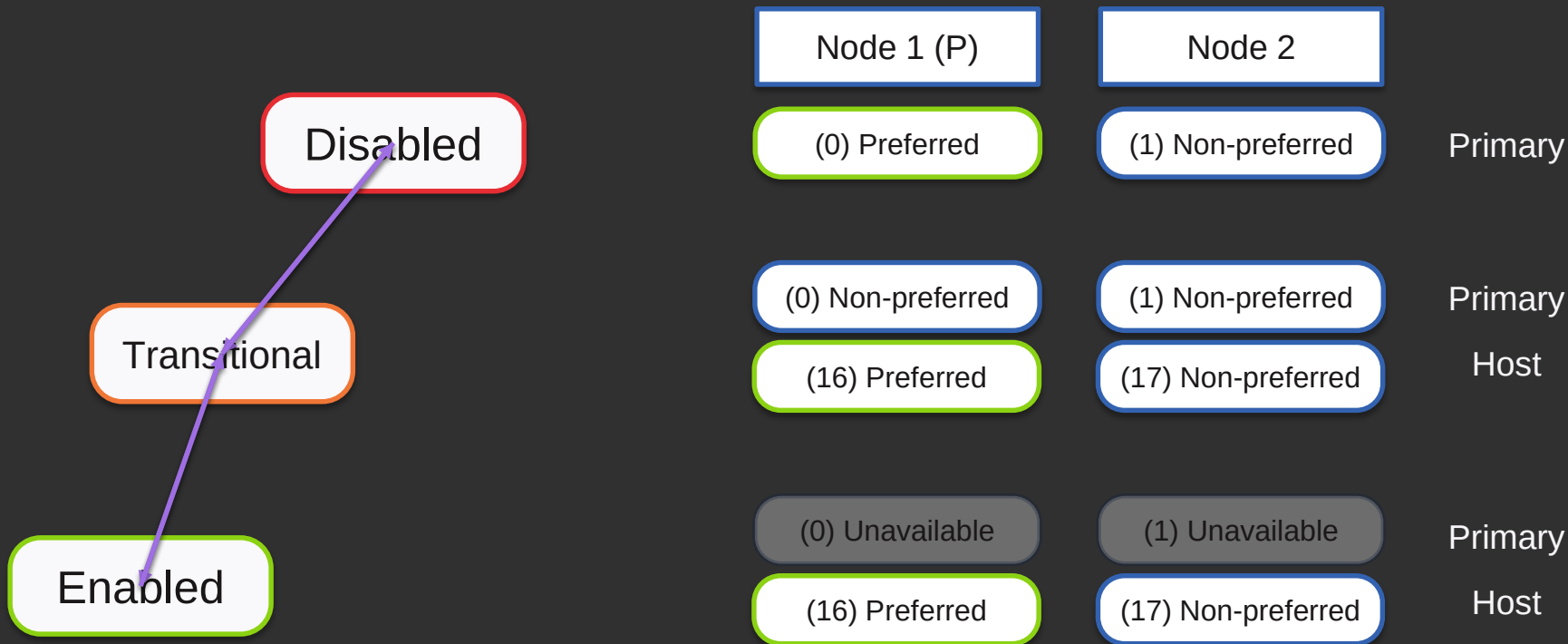
- Confidential slide – removed from distributed version



- **Disabled:**
 - Host ports disabled
 - Host IO through primary ports only
- **Transitional:**
 - Host ports enabled
 - Host IO through primary and host ports
- **Enabled:**
 - Host ports enabled
 - Host IO through host ports only



- Confidential slide – removed from distributed version





- Confidential slide – removed from distributed version



```
>>- swapnode -- -action --+- replace --+->
                                '-- service --'
```

```
>-+-----+-
  '- -permitofflinevolumes --'
```

```
>--+- node id ----+--><
    '-- node name --'
```

swapnode -action replace X
replace offline node X with replacement node

swapnode -action service X
Prepare node X for hardware servicing



- Confidential slide – removed from distributed version



- Pre-7.7 SVC/Storwize nodes cannot use host ports of other clusters
- Host rescan required before disabling the host ports
- Not all host operating systems supported at GA - early PTF aiming to address many of these
- Host ports can only be enabled post-upgrade; cannot be used during upgrades from earlier versions
- swapnode command aimed at early PTF
- FCoE not supported



Discussion